

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

UCI Forecasting Reproducibility Report

1. Environment & Parameters

- **Random Seed:** 42
- **Raw File:** data/uci/LD2011_2014.txt
- **Processed File:** data/processed/uci_hourly_aggregate_load.csv
- **Configuration File:** configs/uci_forecasting_config.yaml
- **Selected Model:** XGBoost Regressor
- **Feature Count:** 29
- **Train Samples:** 26136
- **Validation Samples:** 4344
- **Test Samples:** 4416